05 - Train-Test Split and Validation Strategy

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Separate features and target

```
X = df.drop('Churn', axis=1)
y = df['Churn']
```

X contains predictor columns and y contains the binary outcome. The EDA notebook exports the dummy-encoded dataset to tel_churn.csv; the model notebook reads it and removes the exported dataframe index column named Unnamed: 0.

2. Original train-test split

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2
)
```

The repository uses an 80:20 split. On the cleaned 7,032-row dataset, the notebook reports 1,407 test observations, with the remainder used for training.

3. Recommended split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

random_state makes the split reproducible. stratify=y preserves approximately the same class proportions in both partitions, which is useful for an imbalanced classification task.

4. Critical issue in the repository: resampling before splitting

The model notebook applies SMOTEENN to the entire feature matrix and target first, then splits the resampled data into train and test partitions. This is methodologically incorrect for an unbiased evaluation.

Repository sequence: full dataset → SMOTEENN → train/test split

SMOTE creates synthetic observations using minority-class neighbours; ENN removes observations based on neighbour relationships. If these operations are performed before the split, the eventual test observations can influence synthetic sample generation or cleaning. Related information can therefore cross the training/test boundary, compromising test independence.

5. Correct approach

Correct sequence: original data → train/test split → resample training data only → fit model → evaluate on untouched test data

```
from imblearn.pipeline import Pipeline
from imblearn.combine import SMOTEENN
from sklearn.ensemble import RandomForestClassifier


pipe = Pipeline([
    ("resample", SMOTEENN(random_state=42)),
    ("model", RandomForestClassifier(
```

```
        n_estimators=100, max_depth=6,
        min_samples_leaf=8, random_state=100
    ))
])
pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)
```

An imbalanced-learn Pipeline applies resampling during fit. During cross-validation, each fold's resampling is confined to its training partition. The held-out test dataset remains in its original class distribution and is never resampled.

6. Cross-validation

The repository does not demonstrate cross-validation. A better development process uses StratifiedKFold to evaluate candidate models and tune hyperparameters on training data. The final test set should be held out until model selection and threshold decisions are complete.

7. Interview-ready explanation

> *The notebook uses an 80:20 train-test split. However, it does not set a random state or stratify the target, and more importantly, it applies SMOTEENN before splitting. That is a data-leakage risk because the resampling process sees observations that later enter the test partition. I would correct this by splitting the original data first and placing SMOTEENN and the classifier in an imbalanced-learn pipeline, so resampling happens only inside each training fold.*

8. Questions to prepare

Q. Why do we need a test set?

To estimate generalization on observations not used to fit or select the model.

Q. Why stratify?

To preserve approximate class proportions in each partition.

Q. What is data leakage?

It is the unintended use of information unavailable at prediction time or information from evaluation data during training/model selection, causing unreliable evaluation.

Q. Should the test set be resampled?

No. It should preserve the real-world distribution to estimate performance on the intended population.

Q. Why cross-validate only within training data?

It lets us compare and tune models without repeatedly using the final test set, keeping the test estimate independent.